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Estimating recombination-active defect concentrations in silicon n^+p-p^+ structures via transfer learning from computer vision models

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This study proposes a machine learning framework for quantifying defect concentrations under limited data conditions [1]. Despite the widespread adoption of artificial intelligence in physics, its application to point defect characterization remains limited [2]. Reliable predictive modeling requires large labeled datasets. Simulations partially alleviate this need but may not fully represent physical systems, while exhaustive experiments remain impractical, particularly for semiconductor defect analysis. Transfer learning mitigates this constraint by adapting pretrained models to the target task.

This work investigates the use of pretrained computer vision (CV) models, originally optimized on large-scale datasets, for defect characterization. The approach converts electrophysical measurements governed by defect profiles into image representations compatible with vision architectures, enabling feature extraction and downstream defect analysis. Although CV methods are established for defect detection, existing applications primarily target macroscale defects via electroluminescence imaging [3].

The method was validated by quantifying iron in silicon n^+p-p^+ structures from short-circuit current kinetics measured after FeB dissociation. The training dataset comprised 25 synthetic curves computed for different iron concentrations or 20 experimental curves from samples with varying Fe content. After Savitzky-Golay smoothing, a continuous Morlet wavelet transform mapped the one-dimensional time series to two-dimensional spectrograms, where concentration-dependent variations in kinetic curves produced distinct features (see Fig.1). Feature extraction employed six pretrained CV architectures, yielding 1,000-433,640 features. These vectors were used to train regression models, including RF, SVR, GB, XGB, and DNN, on the aforementioned small datasets.

Fig.2 shows model performance on synthetic and experimental datasets. The most accurate predictions – mean percentage error below 6% for synthetic and 9% for experimental data – were obtained using EfficientNetB7 or NASNetLarge for wavelet images, combined with DNN or SVR. For experimental data, using class probabilities as features was most effective, while direct image features performed best for synthetic datasets.

[1] O. Olikh, O. Zavhorodnii, and Y. Perets, *Semicond. Sci. Technol.* **2026**, *41*, 015021.

[2] Y. Buratti, J. Dick J, Q. Le Gia, and Z. Hameiri, *ACS Appl. Mater. Interfaces* **2022**, *14*, 48647.

[3] R. Jaiswal, M. Martinez-Ramon, and T. Busani, *IEEE J. Photovolt.* **2023**, *13*, 2.

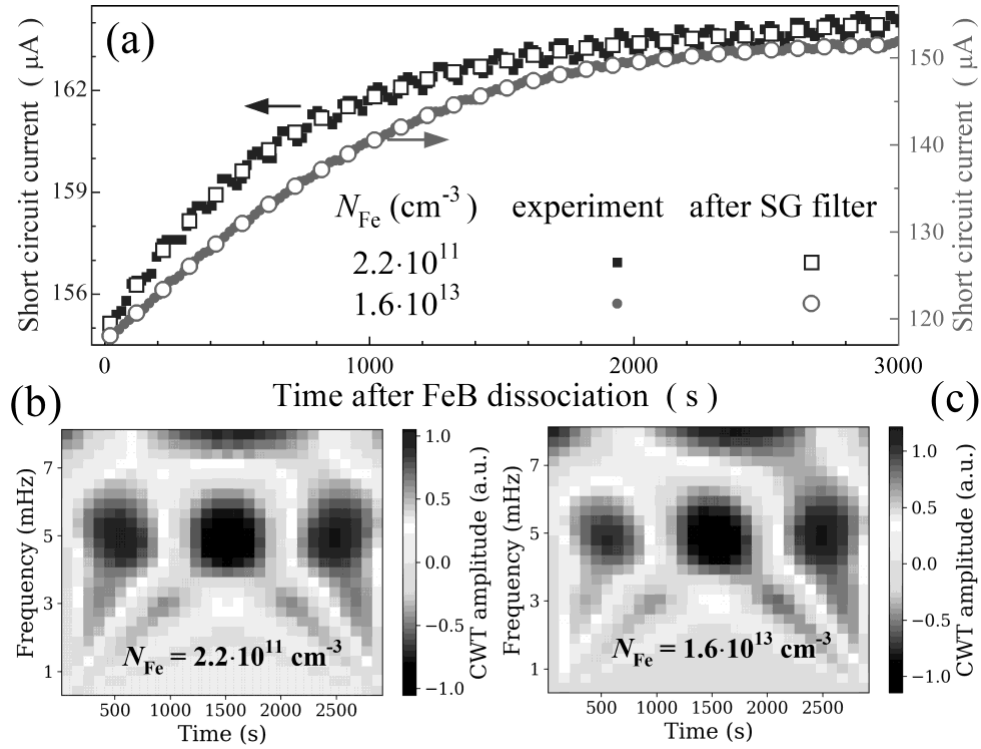


Fig. 1. (a) Short-circuit current kinetics for samples with iron concentration $2.2 \cdot 10^{11} \text{ cm}^{-3}$ (filled squares) and $1.6 \cdot 10^{13} \text{ cm}^{-3}$ (filled circles), with Savitzky–Golay–filtered curves (open symbols). (b, c) Representative wavelet spectrograms from the filtered data. Although the CV models analyzed full-color images, these are shown here in the format compliant with the abstract submission guidelines.

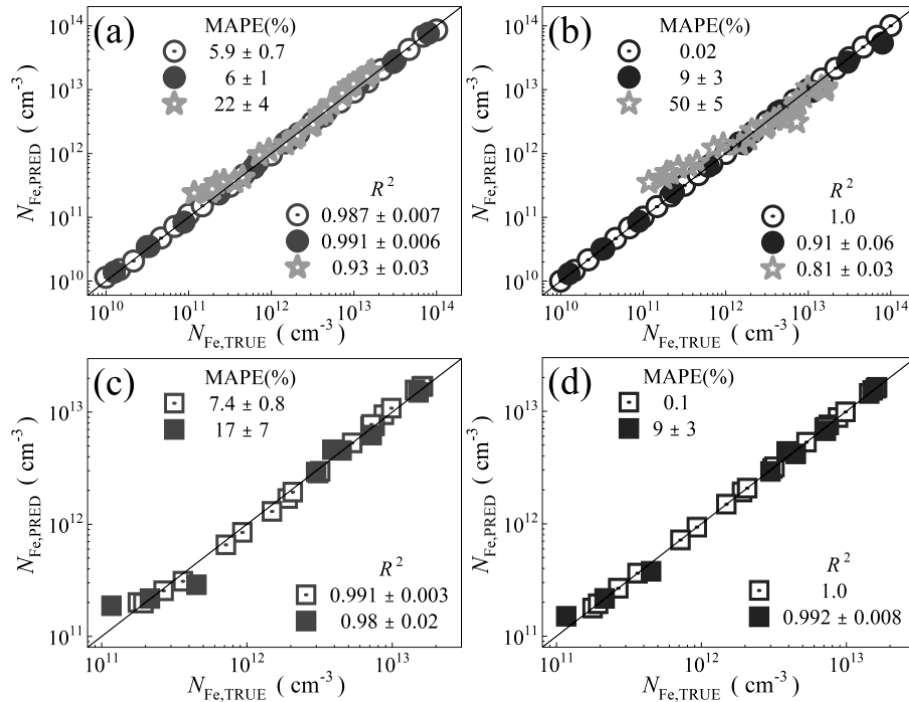


Fig. 2. Comparison of model predictions with ground truth. Models were trained on synthetic (a,b) or experimental (c,d) datasets using EfficientNetB7 with Deep Neural Network (a,c) and NASNetLarge with Support Vector Regression (b,d). Symbols indicate synthetic training (open circles) and testing (filled circles), experimental training (open squares) and testing (filled squares), and stars denote synthetic-trained models applied to experimental data. Mean absolute percentage error (MAPE) and coefficient of determination (R^2) are reported for each case.